US 20250285266A1

(19) United States

(12) Patent Application Publication (10) Pub. No.: US 2025/0285266 A1

Zhao et al. (43) Pub. Date: Sep. 11, 2025

(54) **FLEXIBLE TRANSFORMER FOR MULTIPLE HETEROGENEOUS IMAGE INPUT FOR MEDICAL IMAGING ANALYSIS**

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(21) Appl. No.: **18/961,662**

(22) Filed: **Nov. 27, 2024**

Related U.S. Application Data

(60) Provisional application No. 63/561,959, filed on Mar. 6, 2024.

Publication Classification

(51) **Int. Cl.**

G06T 7/00	(2017.01)
G06V 10/77	(2022.01)
G06V 10/82	(2022.01)

(52) **U.S. Cl.**

CPC ***G06T 7/0012*** (2013.01); ***G06V 10/7715*** (2022.01); ***G06V 10/82*** (2022.01); *G06T 2207/20084* (2013.01)

(57) **ABSTRACT**

Systems and methods for performing a medical imaging analysis task are provided. 1) a plurality of medical images acquired at a plurality of acquisition orientations and in one or more domains and 2) a domain code for each particular acquisition orientation of the plurality of acquisition orientations are received. Each of the domain codes identify a presence of the one or more domains of the plurality of medical images that were acquired at the particular acquisition orientation. For each of the particular acquisition orientations, the domain code for the particular acquisition orientation are encoded, features are extracted from the plurality of medical images that were acquired at the particular acquisition orientation, and the encoded domain code and the extracted features are combined to generate image features for the particular acquisition orientation. A medical imaging analysis task is performed based on the image features for each of the particular acquisition orientations. Results of the medical imaging analysis task are output.

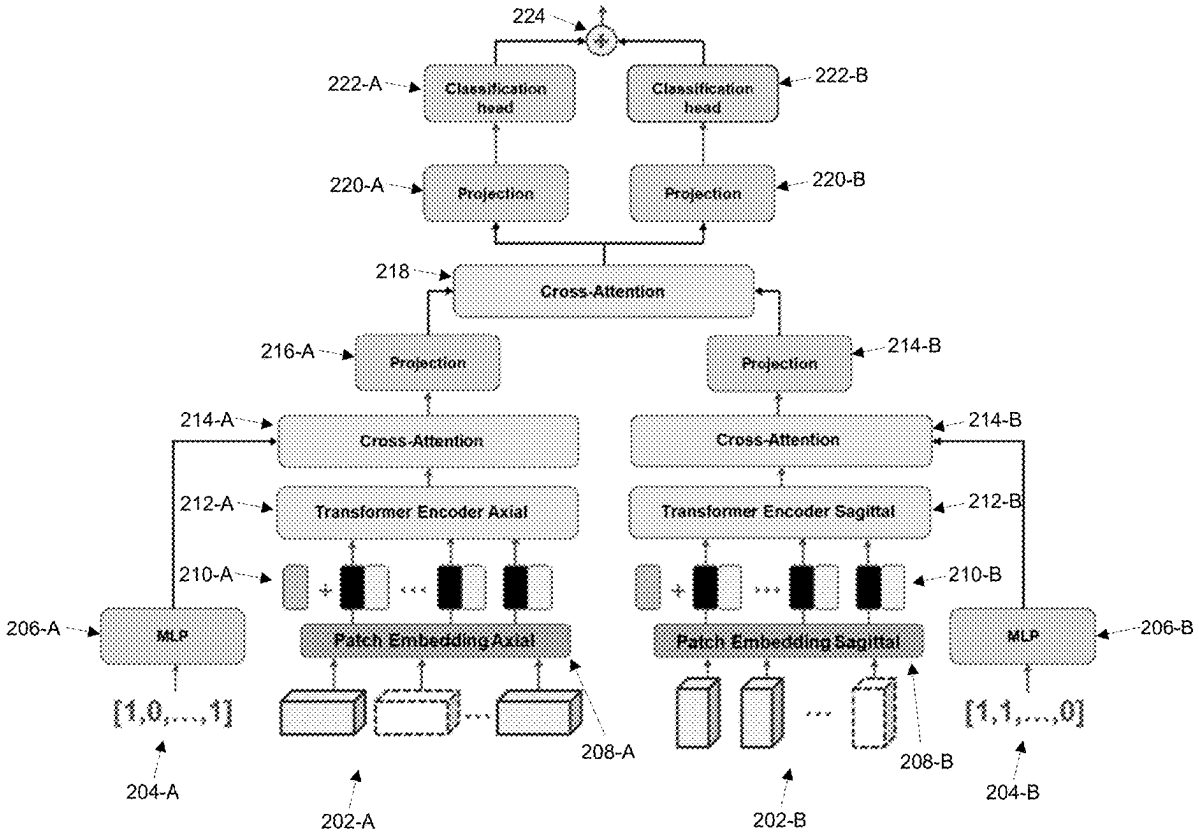
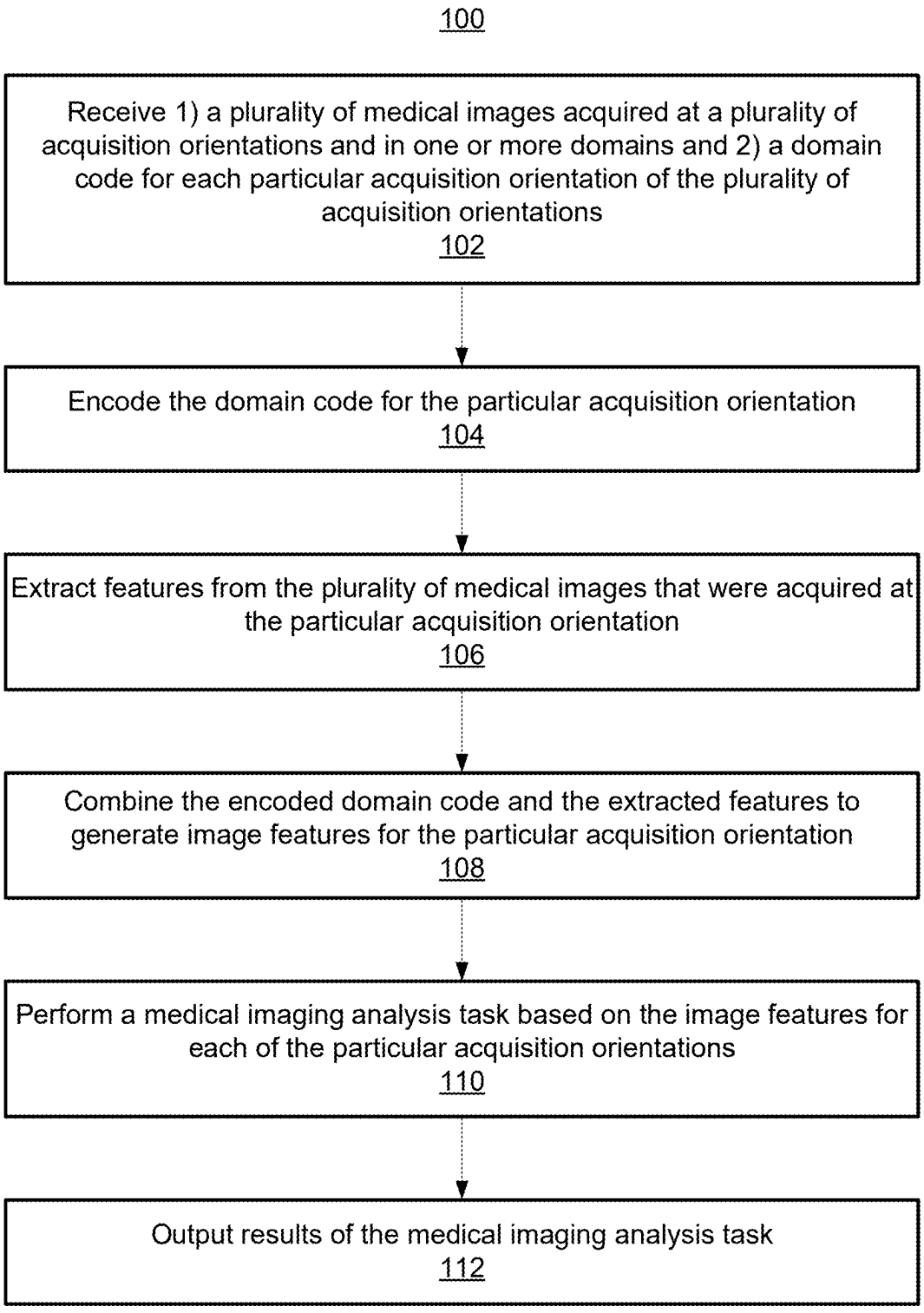


FIG. 1


100
Receive 1) a plurality of medical images acquired at a plurality of acquisition orientations and in one or more domains and 2) a domain code for each particular acquisition orientation of the plurality of acquisition orientations
102
Encode the domain code for the particular acquisition orientation
104
Extract features from the plurality of medical images that were acquired at the particular acquisition orientation
106
Combine the encoded domain code and the extracted features to generate image features for the particular acquisition orientation
108
Perform a medical imaging analysis task based on the image features for each of the particular acquisition orientations
110
Output results of the medical imaging analysis task
112

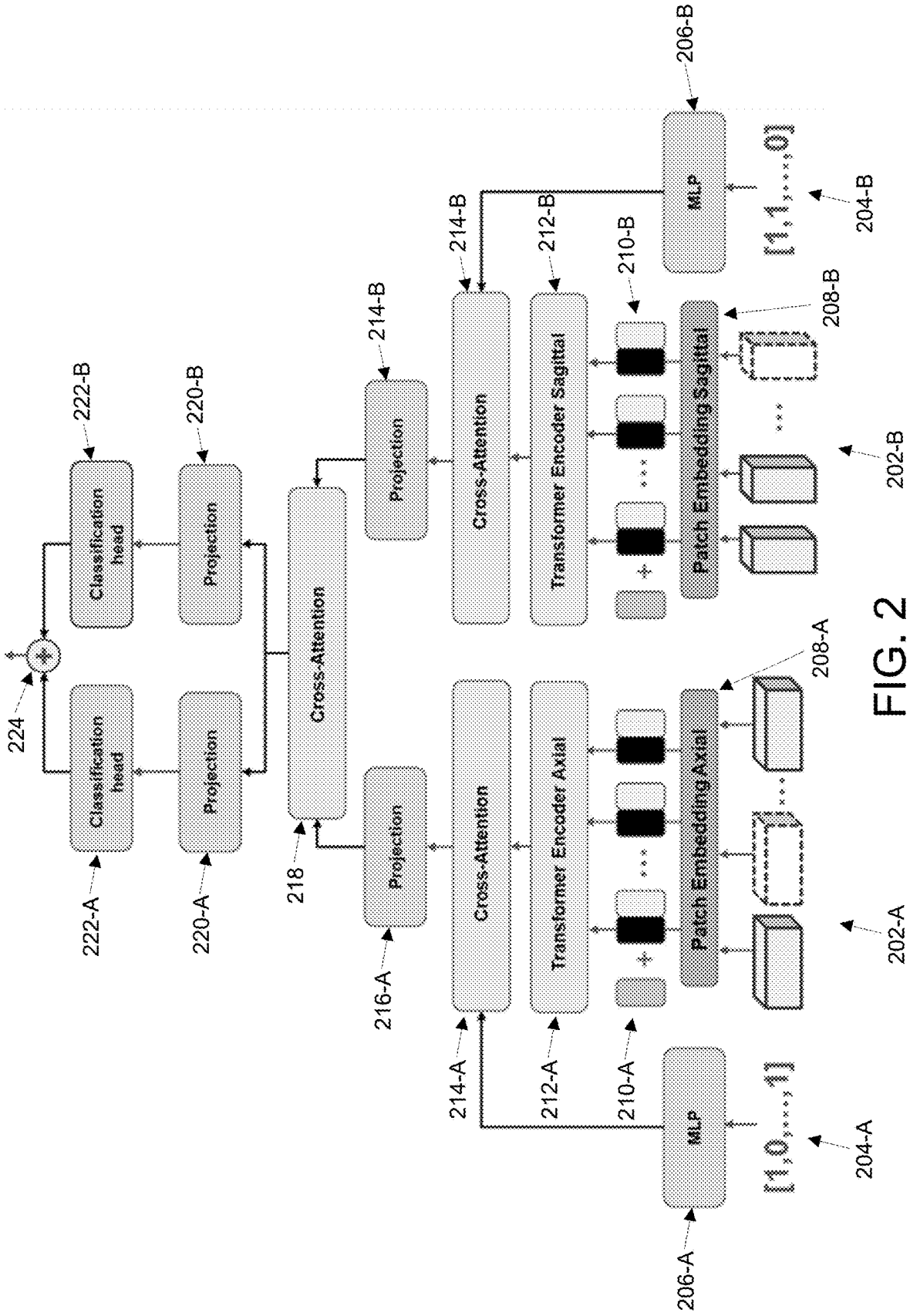
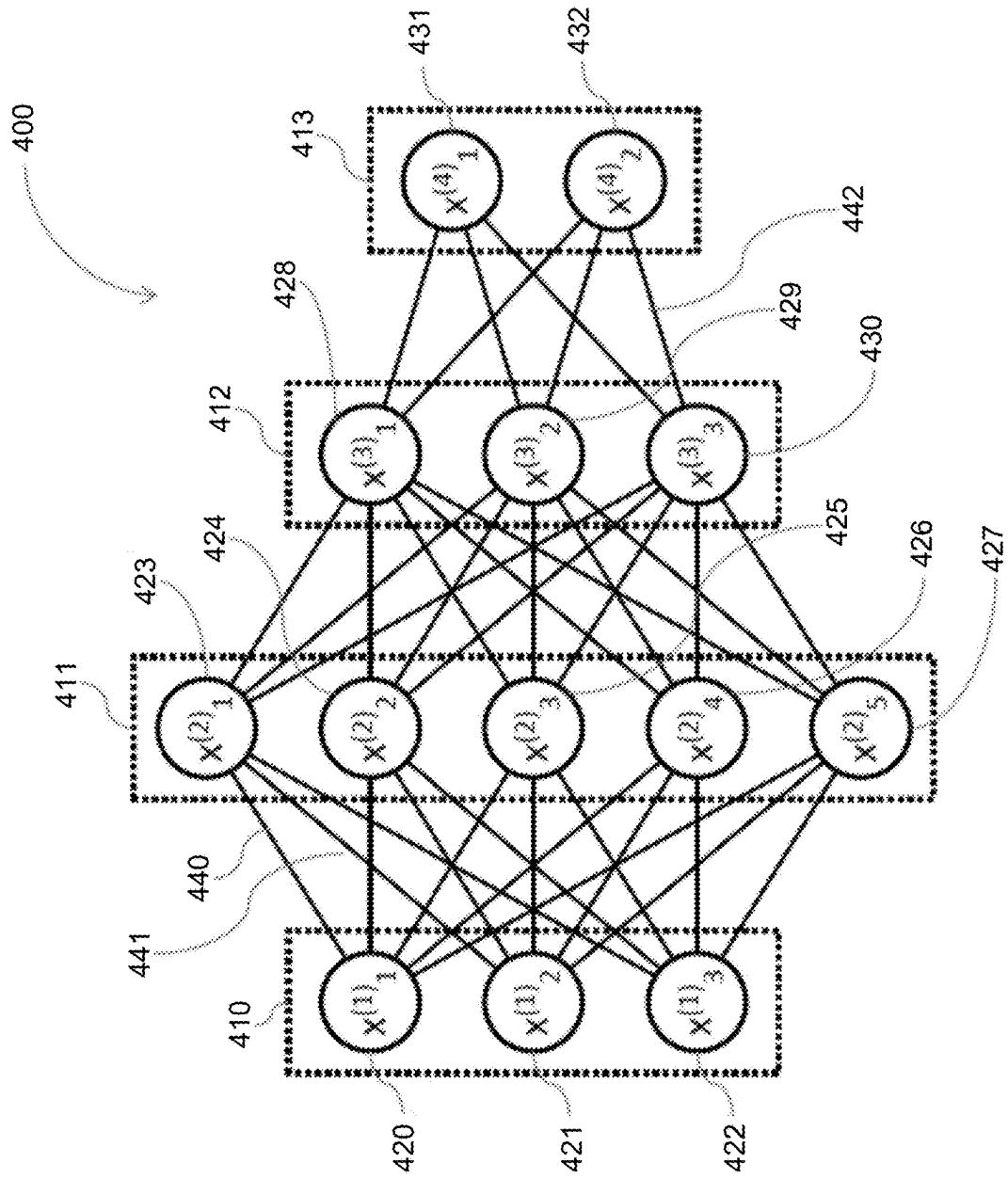

206-A
206-B
MLP
MLP
[1,0,...,1]
[1,1,...,0]
204-A
204-B
208-A
208-B
Patch Embedding Axial
Patch Embedding Sagittal
202-A
202-B
210-A
210-B
212-A
212-B
Transformer Encoder Axial
Transformer Encoder Sagittal
214-A
214-B
Cross-Attention
Cross-Attention
216-A
216-B
Projection
Projection
218
Cross-Attention
220-A
220-B
Projection
Projection
222-A
222-B
Classification head
Classification head
224

FIG. 2

FIG. 3

300

Model	AUC
Proposed	0.8420
DenseNet	0.8363
ResNet	0.8170
ViT	0.7990


400
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440
441
442
$x^{(1)}_1$
$x^{(1)}_2$
$x^{(1)}_3$
$x^{(2)}_1$
$x^{(2)}_2$
$x^{(2)}_3$
$x^{(2)}_4$
$x^{(2)}_5$
$x^{(3)}_1$
$x^{(3)}_2$
$x^{(3)}_3$
$x^{(4)}_1$
$x^{(4)}_2$

FIG. 4

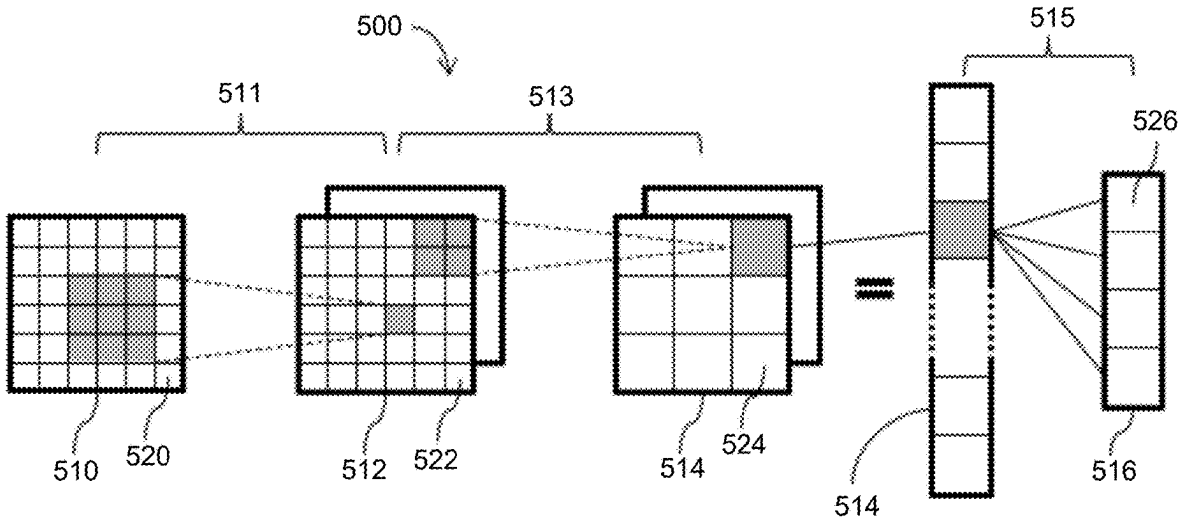


FIG. 5

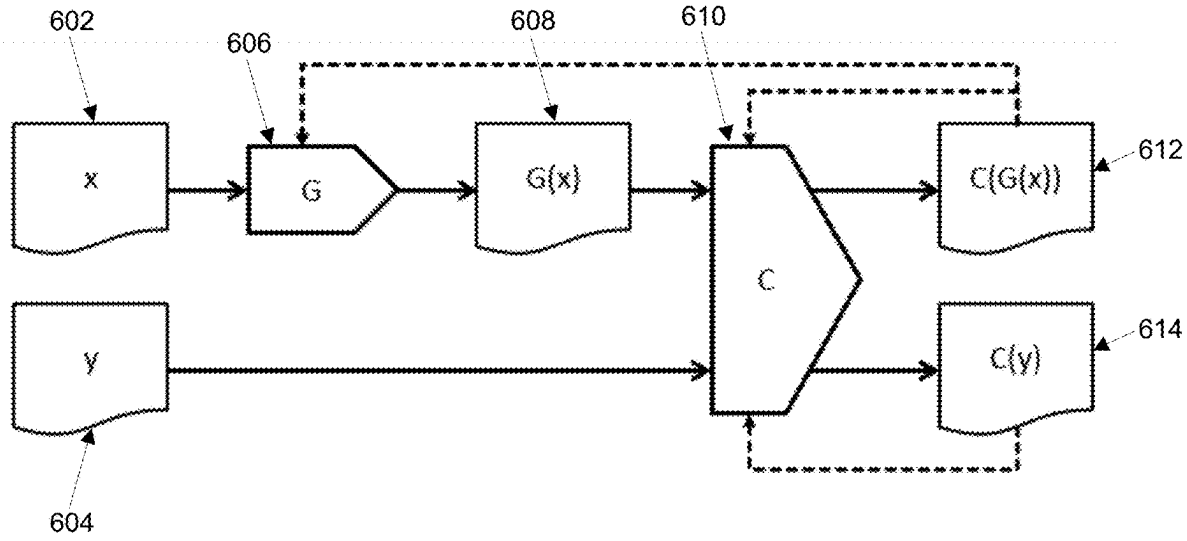


FIG. 6

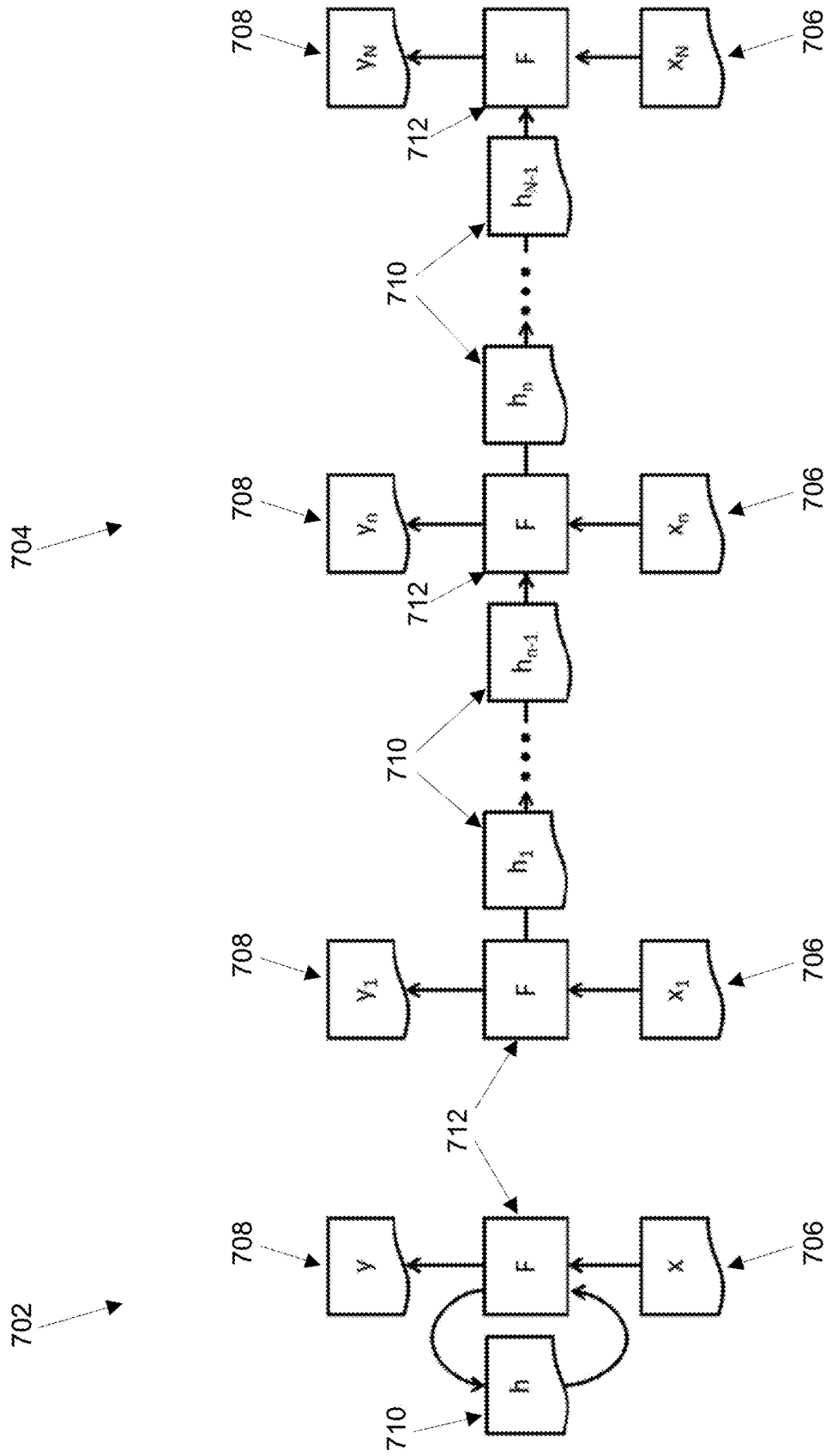
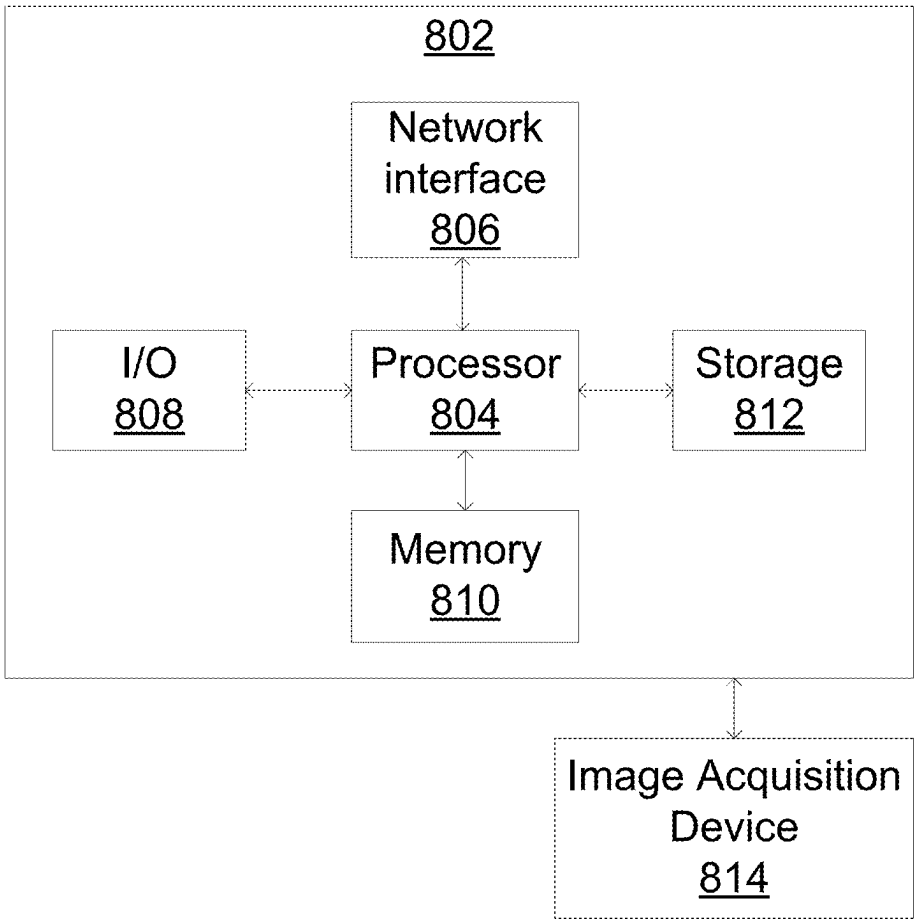


FIG. 7

FIG. 8


802
Network interface 806
I/O 808
Processor 804
Storage 812
Memory 810
Image Acquisition Device 814

FLEXIBLE TRANSFORMER FOR MULTIPLE HETEROGENEOUS IMAGE INPUT FOR MEDICAL IMAGING ANALYSIS

CROSS-REFERENCE TO RELATED APPLICATIONS

[0001] This application claims the benefit of U.S. Provisional Application No. 63/561,959, filed Mar. 6, 2024, the disclosure of which is herein incorporated by reference in its entirety.

TECHNICAL FIELD

[0002] The present invention relates generally to AI/ML (artificial intelligence/machine learning) based medical imaging analysis, and in particular to a flexible transformer for multiple heterogeneous image input for medical imaging analysis.

BACKGROUND

[0003] In medical imaging, the acquisition of medical images often entails diverse protocols across various clinical sites, resulting in significant cross-patient and in-patient heterogeneity within medical imaging datasets. Notably, in MRI (magnetic resonance imaging) acquisition, a single imaging session typically yields multiple 3D images characterized by multiple contrasts, different acquisition orientations, and varying anisotropic resolutions. Such broad heterogeneity in medical imaging datasets introduces complexity in utilizing these images for analysis using neural networks. Conventional approaches for medical imaging analysis using neural networks typically involve resampling or resizing images to uniform resolutions or sizes. However, such conventional approaches risk information loss, particularly for anisotropic images with different orientations. In addition, preprocessing pipelines for resampling or resizing images for each clinical application typically increases development costs.

BRIEF SUMMARY OF THE INVENTION

[0004] In accordance with one or more embodiments, systems and methods for performing a medical imaging analysis task are provided. 1) a plurality of medical images acquired at a plurality of acquisition orientations and in one or more domains and 2) a domain code for each particular acquisition orientation of the plurality of acquisition orientations are received. Each of the domain codes identify a presence of the one or more domains of the plurality of medical images that were acquired at the particular acquisition orientation. For each of the particular acquisition orientations, the domain code for the particular acquisition orientation are encoded, features are extracted from the plurality of medical images that were acquired at the particular acquisition orientation, and the encoded domain code and the extracted features are combined to generate image features for the particular acquisition orientation. A medical imaging analysis task is performed based on the image features for each of the particular acquisition orientations. Results of the medical imaging analysis task are output.

[0005] In one embodiment, a first set of features is extracted from the input tensor of that acquisition orientation using a first machine learning based encoder. The first set of features are encoded into a second set of features using a second machine learning based encoder to generate the extracted features. In one embodiment, the first set of features is encoded with positional embeddings. In one embodiment, the first set of features are lower level features and the second set of features are higher level features.

[0006] In one embodiment, the plurality of medical images that were acquired at the particular acquisition orientation is resampled onto a same pixel grid for that particular acquisition orientation to form aligned images. The aligned images are combined to form an input tensor. The first set of features is extracted from the input tensor using the first machine learning based encoder.

[0007] In one embodiment, the encoded domain code and the extracted features are combined using a cross-attention layer.

[0008] In one embodiment, the image features for each of the particular acquisition orientations are combined to generate combined features. The medical imaging analysis task is performed based on the combined features.

[0009] In one embodiment, each of the domain codes identify an absence of certain domains of a set of predefined domains from the one or more domains of the plurality of medical images that were acquired at the particular acquisition orientation.

[0010] In one embodiment, the plurality of acquisition orientations comprises at least one of axial, sagittal, or coronal.

[0011] These and other advantages of the invention will be apparent to those of ordinary skill in the art by reference to the following detailed description and the accompanying drawings.

BRIEF DESCRIPTION OF THE DRAWINGS

[0012] FIG. **1** shows a method for assessing for performing a medical imaging analysis task on a plurality of medical images acquired at a plurality of acquisition orientations, in accordance with one or more embodiments;

[0013] FIG. **2** shows a network architecture of a transformer for performing a medical imaging analysis task on a plurality of medical images acquired at a plurality of acquisition orientations, in accordance with one or more embodiments;

[0014] FIG. **3** shows a table comparing performance of the transformer model in accordance with embodiments described herein with conventional models;

[0015] FIG. **4** shows an exemplary artificial neural network that may be used to implement one or more embodiments;

[0016] FIG. **5** shows a convolutional neural network that may be used to implement one or more embodiments;

[0017] FIG. **6** shows a data flow diagram of a generative adversarial network that may be used to implement one or more embodiments;

[0018] FIG. **7** shows a schematic structure of a recurrent machine learning model that may be used to implement one or more embodiments; and

[0019] FIG. **8** shows a high-level block diagram of a computer that may be used to implement one or more embodiments.

DETAILED DESCRIPTION

[0020] The present invention generally relates to methods and systems for medical imaging analysis using a flexible transformer for multiple heterogeneous image input.

Embodiments of the present invention are described herein to give a visual understanding of such methods and systems. A digital image is often composed of digital representations of one or more objects (or shapes). The digital representation of an object is often described herein in terms of identifying and manipulating the objects. Such manipulations are virtual manipulations accomplished in the memory or other circuitry/hardware of a computer system. Accordingly, is to be understood that embodiments of the present invention may be performed within a computer system using data stored within the computer system. Further, reference herein to pixels of an image may refer equally to voxels of an image and vice versa.

[0021] Embodiments described herein provide for a novel flexible transformer architecture designed to effectively manage multiple anisotropic input medical images acquired at different acquisition orientations. The transformer architecture eliminates the necessity for extensive resampling or resizing the input medical images, thereby preserving the integrity of the images and preventing loss of information. By circumventing the preprocessing steps for resampling or resizing, the transformer architecture optimizes the performance and robustness of deep learning models specifically tailored for performing medical imaging analysis tasks.

[0022] FIG. **1** shows a method **100** for performing a medical imaging analysis task on a plurality of medical images acquired at a plurality of acquisition orientations, in accordance with one or more embodiments. The steps and sub-steps of method **100** may be performed by one or more suitable computing devices, such as, e.g., computer **802** of FIG. **8**. FIG. **2** shows a network architecture **200** of a transformer for performing a medical imaging analysis task on a plurality of medical images acquired at a plurality of acquisition orientations, in accordance with one or more embodiments. FIG. **1** and FIG. **2** will be described together.

[0023] At step **102** of FIG. **1**, 1) a plurality of medical images acquired at a plurality of acquisition orientations and in one or more domains and 2) a domain code for each particular acquisition orientation of the plurality of acquisition orientations are received. Each of the domain codes identify a presence of the one or more domains of the plurality of medical images that were acquired at the particular acquisition orientation. In one example, as shown in network architecture **200** of FIG. **2**, the plurality of medical images may be 3D medical images **202**-A acquired at an axial orientation of a patient and medical images **202**-B acquired at a sagittal orientation of the patient and the domain codes may be domain code **204**-A identifying one or more domains of medical images **202**-A and domain code **204**-B identifying one or more domains of medical images **202**-B (medical images **202**-A and **202**-B are collectively referred to as medical images **202** and domain codes **204**-A and **204**-B are collectively referred to as domain codes **204**).

[0024] The plurality of medical images may depict any anatomical object of interest of a patient at the plurality of acquisition orientations. The plurality of acquisition orientations may comprise, for example, axial, sagittal, coronal, or any other suitable acquisition orientation of the patient. As used herein, a domain of a medical image refers to the modality of the medical image as well as the protocol used for obtaining the medical image in that modality. The modality of the one or more input medical images may include, for example, MRI (magnetic resonance imaging), CT (computed tomography), US (ultrasound), x-ray, SPECT (single-photon emission computed tomography), PET (positron emission tomography), or any other medical imaging modality or combinations of medical imaging modalities. The protocol used for obtaining the medical image may include, for example, acquisition sequences or techniques for acquiring a medical image, such as, e.g., T1-weighted, T2-weighted, proton density-weighted MRI images, contrast and non-contrast images, CT images captured with low kV (kilovoltage) and high kV, or low and high resolution medical images. Accordingly, the domains may be completely different medical imaging modalities or different image protocols within the same overall imaging modality. The plurality of medical images may be represented in the image space (e.g., as pixel or voxel values in spatial coordinates) or the latent space (e.g., as a lower-dimensional, compressed representation of the one or more medical images represented as a feature vector). The plurality of medical images in the image space may be 3D (three dimensional) volumes.

[0025] The domain code for each particular acquisition orientation identifies a presence of the one or more domains of the plurality of medical images that were acquired at the particular acquisition orientation for a set of predefined domains, or an absence of certain domains of the set of predefined domains from the one or more domains of the plurality of medical images. The set of predefined domains represents the domains that machine learning based encoders (utilized at step **106**) are trained to process. The domain code may be represented in any suitable form for identifying the presence of the one or more domains in the set of predefined domains and/or the absence of the certain domains of the set of predefined domains from the one or more domains. In one embodiment, the domain codes are $1 \times n$ vectors, where n represents the number of domains in the set of predefined domains. Each position in the vector is associated with a respective domain of the set of predefined domains. The value at each position may be defined, for example, by a one-hot, where a value of 1 defines the presence of an input medical image in the associated domain and a value of 0 defines the absence of an input medical image in the associated domain. Other approaches for encoding the presence and/or absence of domains in the set of predefined domains are also contemplated.

[0026] The one or more medical images and/or the domain code may be received, for example, by directly receiving the one or more medical images from an image acquisition device (e.g., image acquisition device **814** of FIG. **8**) as the one or more medical images are acquired, by loading the one or more medical images and/or the domain code from a storage or memory of a computer system (e.g., storage **812** or memory **810** of computer **802** of FIG. **8**), or by receiving the one or more medical images and/or the domain code from a remote computer system (e.g., computer **802** of FIG. **8**). Such a computer system or remote computer system may comprise one or more patient databases, such as, e.g., an EHR (electronic health record), EMR (electronic medical record), PHR (personal health record), HIS (health information system), RIS (radiology information system), PACS (picture archiving and communication system), LIMS (laboratory information management system), or any other suitable database or system.

[0027] Method **100** of FIG. **1** proceeds to steps **104**-**108**, which are performed for each of the particular acquisition orientations of the plurality of acquisition orientations.

[0028] At step **104** of FIG. **1**, the domain code for the particular acquisition orientation is encoded. In one embodiment, the domain code is encoded using an MLP (multilayer perceptron). For example, in one example, as shown in network architecture **200** of FIG. **2**, domain codes **204**-A and **204**-B are respectively encoded by MLP **206**-A and **206**-B (collectively referred to as MLPs **206**). While MLP **206**-A and **206**-B are separately shown in network architecture **200** to illustrate processing of domain codes **204**-A and **204**-B, it should be understood that MLP **206**-A and **206**-NB are the same MLP. The MLP receives as input the domain code for the particular acquisition orientation, transforms the domain code into features or embeddings representing a lower-dimensional, compressed representation of the domain code, and outputs the features. The domain code for the particular acquisition orientation may be encoded using any other suitable approach. For example, the domain code for the particular acquisition orientation may be encoded using a learnable linear projector that maps the domain code to the features through a set of parameters optimized during the training process.

[0029] At step **106** of FIG. **1**, features are extracted from the plurality of medical images that were acquired at the particular acquisition orientation. In one embodiment, a first set of lower-level features (tokens) are extracted from the plurality of medical images that were acquired at the particular acquisition orientation and the first set of features are encoded into a second set of higher-level features.

[0030] The first set of features are extracted from the plurality of medical images that were acquired at the particular acquisition orientation using a first machine learning based encoder. The first machine learning based encoder serves as the tokenizer, and it may be, for example, a patch embedding layer, which may be implemented, for example, as a CNN (convolutional neural network). In one example, as shown in network architecture **200** of FIG. **2**, features **210**-A and **210**-B (collectively referred to as features **210**) are extracted from medical images **202**-A and **202**-B using patch embedding layers **208**-A and **208**-B (collectively referred to as patch embedding layers **208**), respectively. However, the first machine learning based encoder may be implemented according to any other suitable machine learning based architecture. The first machine learning based encoder is trained to extract the first set of features from medical images acquired at the particular acquisition orientation. For example, patch embedding layer **208**-A is trained to extract features from medical images acquired at an axial orientation and patch embedding layer **208**-B is trained to extracted features from medical images acquired at a sagittal orientation.

[0031] The plurality of medical images that were acquired at each particular acquisition orientation are resampled onto the same pixel grid for that particular acquisition orientation to form aligned 3D images. For example, the image grid may be a (256, 256, 32) grid for medical images acquired at the axial orientation or a (256, 32, 256) grid for medical images acquired at the sagittal orientation. Then the aligned 3D images are combined (e.g., concatenated) along the channel dimension to form an input tensor for each acquisition orientation. The input tensor is divided into many 3D patches. The first machine learning based encoder receives as input the 3D patches of the plurality of medical images and generates as output the first set of features. The features of the first set of features are lower-dimensional, compressed representations of 3D patches in the 3D volumes represented as feature vectors (tokens). The first machine learning based encoder encodes the patches of the input tensor to features while capturing and encoding the most important features.

[0032] In one embodiment, the first set of features are encoded with positional embeddings (e.g., sinusoidal or learnable). The position embeddings may be fixed or trainable. For example, the position embeddings may be derived based on the spatial location of the 3D patches from which the features were extracted. Specifically, the position embedding matching ensures consistency in position embeddings across features of the different acquisition orientations. Tokens originating from features of different acquisition orientations, each representing corresponding spatial locations within differently oriented images, are paired with identical position embeddings. In one embodiment, learnable position embeddings are utilized, allowing for the dynamic adaptation of position embeddings to capture relative spatial relationships among patches from distinct branches. The learnable position embeddings enhance the ability to effectively learn and integrate information across different acquisition orientations within multi-domain medical images.

[0033] The first set of features are encoded into the second set of features using a second machine learning based encoder. The second machine learning based encoder may be a transformer encoder. In one example, as shown in network architecture **200** of FIG. **2**, features **210**-A and **210**-B are respectively encoded into the second set of features using transformer encoder **212**-A and **212**-B (collectively referred to as transformer encoders **212**). The second machine learning based encoder is trained to extract the second set of features from medical images acquired at the particular acquisition orientation. For example, transformer encoder **210**-A is trained to extract features from medical images acquired at an axial orientation and transformer encoder **210**-B is trained to extracted features from medical images acquired at a sagittal orientation. However, the second machine learning based encoder may be implemented according to any other suitable machine learning based architecture. The second machine learning based encoder receives as input the first set of features and generates as output the second set of features. Compared with the first set of features, the second set of features are higher level of features.

[0034] At step **108** of FIG. **1**, the encoded domain code and the extracted features are combined to generate image features for the particular acquisition orientation. In one embodiment, the encoded domain code and the extracted features are combined using a cross-attention layer. For example, as shown in network architecture **200** of FIG. **2**, the encoded domain code output from MLP **206**-A and **206**-B and the extracted features output by transformer encoder **212**-A and **212**-B are respectively combined by cross-attention layer **214**-A and **214**-B (collectively referred to as cross-attention layers **214**). The cross-attention layer enables the encoded domain code to attend to the extracted features (or vice versa) via query, key, and value representation, which facilitates the extraction and integration of pertinent information from the encoded domain code and the extracted features.

[0035] At step **110** of FIG. **1**, a medical imaging analysis task is performed based on the image features for each of the particular acquisition orientations. In one embodiment, the

image features for each of the particular acquisition orientations are first combined to generate combined features for the plurality of acquisition orientations and the medical imaging analysis task is performed based on the combined features.

[0036] The image features for each of the particular acquisition orientations may be combined using projection layers and a cross-attention layer. For example, as shown in network architecture **200** of FIG. **2**, the image features output by cross-attention layers **214**-A and **214**-B are respectively projected in a higher dimension by projections layers **216**-A and **216**-B (collectively referred to as projection layers **216**) and the projected image features are combined by cross-attention layer **218**. The cross-attention layer enables the image features for one particular acquisition orientation to attend to the image features of the other particular acquisition orientations via query, key, and value representation, which facilitates the extraction and integration of pertinent information from the particular acquisition orientations.

[0037] The medical imaging analysis task is then performed based on the combined features using a projection layer and a task specific machine learning based decoder (e.g., a transformer decoder) for each particular acquisition orientation. For example, as shown in network architecture **200** of FIG. **2**, the combined features are respectively projected into a higher dimension by projection layers **220**-A and **220**-B (collectively referred to as projection layers **220**), classification heads **222**-A and **222**-B (collectively referred to as classification heads **222**) determine a classification based on the projected combined features, and the classifications are combined via combiner **224** to generate a final classification. Projection layers **220**-A and **220**-B and classification heads **222**-A and **222**-B are respectively trained to process the combined features from medical images acquired at the particular acquisition orientation.

[0038] In one embodiment, the medical imaging analysis task is image synthesis for generating a synthetic image from the plurality of medical images. The synthetic image may be in a domain that is absent from the one or more domains of the plurality of medical images. The medical imaging analysis task may additionally or alternatively comprise any other suitable task, such as, e.g., detection, segmentation, classification, quantification, etc.

[0039] At step **112** of FIG. **1**, results of the medical imaging analysis task are output. For example, the results of the medical imaging analysis task can be output by displaying the results on a display device of a computer system (e.g., I/O **808** of computer **802** of FIG. **8**), storing the results on a memory or storage of a computer system (e.g., memory **810** or storage **812** of computer **802** of FIG. **8**) or by transmitting the results to a remote computer system (e.g., computer **802** of FIG. **8**).

[0040] While network architecture **200** of FIG. **2** is implemented with two branches (i.e., a branch for processing medical images **202**-A acquired at an axial orientation and a branch for processing medical images **202**-B acquired at a sagittal orientation), it should be understood that network architecture **200** may be implemented with any number of branches for processing medical images acquired at any number of acquisition orientations. For example, in one embodiment, network architecture **200** may be implemented with three branches to process medical images acquired in axial and sagittal orientations (as shown in FIG. **2**), as well as a coronal orientation.

[0041] In one embodiment, network architecture **200** may be implemented to process medical images acquired in similar orientations but at different native resolutions, e.g., to combine high-resolution structural images (e.g., T1w, T2w, FLAIR (fluid attenuated inversion recovery)) and low-resolution quantitative/metabolic/function information (e.g., DWI (diffusion-weighted imaging), fMRI (functional MRI), first-pass perfusion, MRSI (magnetic resonance spectroscopic imaging)). In this embodiment, the outputs from each branch (output from projection layers **216**) are combined at cross-attention layer **218** with a two-way or three-way cross-attention. Queries are utilized from one particular branch and keys and values from the other branches are utilized to calculate the output from the particular branch providing the queries. By iteratively applying this procedure across all the branches, the outputs of each branch are obtained as the input for subsequent stages of the network for performing the medical imaging analysis task. This approach ensures comprehensive integration of information from multiple orientations or resolutions, enhancing the transformer's capacity to effectively process diverse multi-domain medical images.

[0042] FIG. **3** shows a table **300** comparing performance of the transformer model in accordance with embodiments described herein with conventional models. Table **300** shows results on brain hemorrhage classification with multiple anisotropic MR images acquired with different acquisition orientations, including axial FLAIR, ADC (apparent diffusion coefficient), Trace, GRE (gradient echo sequences), SWI (susceptibility weighted imaging), T2, and sagittal T2 3D MR images on a testing dataset with **144** positive and **1352** negative cases. As shown in table **300**, the proposed transformer model in accordance with embodiments described herein has an AUC (area under the curve) of 0.8420, outperforming DenseNet, ResNet and ViT (vision transformer).

[0043] Embodiments described herein are described with respect to the claimed systems as well as with respect to the claimed methods. Features, advantages or alternative embodiments herein can be assigned to the other claimed objects and vice versa. In other words, claims and embodiments for the systems can be improved with features described or claimed in the context of the respective methods. In this case, the functional features of the method are implemented by physical units of the system.

[0044] Furthermore, certain embodiments described herein are described with respect to methods and systems utilizing trained machine learning models, as well as with respect to methods and systems for providing trained machine learning models. Features, advantages or alternative embodiments herein can be assigned to the other claimed objects and vice versa. In other words, claims and embodiments for providing trained machine learning models can be improved with features described or claimed in the context of utilizing trained machine learning models, and vice versa. In particular, datasets used in the methods and systems for utilizing trained machine learning models can have the same properties and features as the corresponding datasets used in the methods and systems for providing trained machine learning models, and the trained machine learning models provided by the respective methods and

systems can be used in the methods and systems for utilizing the trained machine learning models.

[0045] In general, a trained machine learning model mimics cognitive functions that humans associate with other human minds. In particular, by training based on training data the machine learning model is able to adapt to new circumstances and to detect and extrapolate patterns. Another term for "trained machine learning model" is "trained function."

[0046] In general, parameters of a machine learning model can be adapted by means of training. In particular, supervised training, semi-supervised training, unsupervised training, reinforcement learning and/or active learning can be used. Furthermore, representation learning (an alternative term is "feature learning") can be used. In particular, the parameters of the machine learning models can be adapted iteratively by several steps of training. In particular, within the training a certain cost function can be minimized. In particular, within the training of a neural network the backpropagation algorithm can be used.

[0047] In particular, a machine learning model, such as, e.g., the MLP utilized at step **104**, the first and second machine learning based encoders utilized at step **106**, the cross-attention layer utilized at step **108**, and/or the cross-attention layer or the machine learning based decoders utilized at step **110** of FIG. **1**, or MLPs **206**, patch embedding layers **208**, transformer encoders **212**, cross-attention layers **214**, projection layers **216**, cross-attention layer **218**, projection layers **220**, and/or classification heads **222** of FIG. **2**, can comprise, for example, a neural network, a support vector machine, a decision tree and/or a Bayesian network, and/or the machine learning model can be based on, for example, k-means clustering, Q-learning, genetic algorithms and/or association rules. In particular, a neural network can be, e.g., a deep neural network, a convolutional neural network or a convolutional deep neural network. Furthermore, a neural network can be, e.g., an adversarial network, a deep adversarial network and/or a generative adversarial network.

[0048] FIG. **4** shows an embodiment of an artificial neural network **400** that may be used to implement one or more machine learning models described herein. Alternative terms for "artificial neural network" are "neural network", "artificial neural net" or "neural net".

[0049] The artificial neural network **400** comprises nodes **420**, . . . , **432** and edges **440**, **442**, wherein each edge **440**, . . . , **442** is a directed connection from a first node **420**, **432** to a second node **420**, . . . , **432**. In general, the first node **420**, . . . , **432** and the second node **420**, . . . , **432** are different nodes **420**, . . . , **432**, it is also possible that the first node **420**, . . . , **432** and the second node **420**, . . . , **432** are identical. For example, in FIG. **4** the edge **440** is a directed connection from the node **420** to the node **423**, and the edge **442** is a directed connection from the node **430** to the node **432**. An edge **440**, . . . , **442** from a first node **420**, . . . , **432** to a second node **420**, . . . , **432** is also denoted as "ingoing edge" for the second node **420**, . . . , **432** and as "outgoing edge" for the first node **420**, . . . **432**.

[0050] In this embodiment, the nodes **420**, . . . , **432** of the artificial neural network **400** can be arranged in layers **410**, . . . , **413**, wherein the layers can comprise an intrinsic order introduced by the edges **440**, . . . , **442** between the nodes **420**, . . . , **432**. In particular, edges **440**, . . . , **442** can exist only between neighboring layers of nodes. In the displayed embodiment, there is an input layer **410** comprising only nodes **420**, . . . , **422** without an incoming edge, an output layer **413** comprising only nodes **431**, **432** without outgoing edges, and hidden layers **411**, **412** in-between the input layer **410** and the output layer **413**. In general, the number of hidden layers **411**, **412** can be chosen arbitrarily. The number of nodes **420**, . . . , **422** within the input layer **410** usually relates to the number of input values of the neural network, and the number of nodes **431**, **432** within the output layer **413** usually relates to the number of output values of the neural network.

[0051] In particular, a (real) number can be assigned as a value to every node **420**, . . . , **432** of the neural network **400**. Here, $x^{(n)}_i$ denotes the value of the i-th node **420**, . . . , **432** of the n-th layer **410**, . . . , **413**. The values of the nodes **420**, . . . , **422** of the input layer **410** are equivalent to the input values of the neural network **400**, the values of the nodes **431**, **432** of the output layer **413** are equivalent to the output value of the neural network **400**. Furthermore, each edge **440**, . . . , **442** can comprise a weight being a real number, in particular, the weight is a real number within the interval [−1, 1] or within the interval [0, 1]. Here, $w^{(m,n)}_{i,j}$ denotes the weight of the edge between the i-th node **420**, . . . , **432** of the m-th layer **410**, . . . , **413** and the j-th node **420**, . . . , **432** of the n-th layer **410**, . . . , **413**. Furthermore, the abbreviation $w^{(n)}_{i,j}$ is defined for the weight $w^{(n,n+1)}_{i,j}$.

[0052] In particular, to calculate the output values of the neural network **400**, the input values are propagated through the neural network. In particular, the values of the nodes **420**, . . . , **432** of the (n+1)-th layer **410**, . . . , **413** can be calculated based on the values of the nodes **420**, . . . , **432** of the n-th layer **410**, . . . , **413** by

$$x^{(n+1)}_j = f\left(\sum_i x^{(n)}_i \cdot w^{(n)}_{i,j}\right).$$

[0053] Herein, the function f is a transfer function (another term is "activation function"). Known transfer functions are step functions, sigmoid function (e.g., the logistic function, the generalized logistic function, the hyperbolic tangent, the Arctangent function, the error function, the smoothstep function) or rectifier functions. The transfer function is mainly used for normalization purposes.

[0054] In particular, the values are propagated layer-wise through the neural network, wherein values of the input layer **410** are given by the input of the neural network **400**, wherein values of the first hid-den layer **411** can be calculated based on the values of the input layer **410** of the neural network, wherein values of the second hidden layer **412** can be calculated based in the values of the first hidden layer **411**, etc.

[0055] In order to set the values $w^{(m,n)}_{i,j}$ for the edges, the neural network **400** has to be trained using training data. In particular, training data comprises training input data and training output data (denoted as t_i). For a training step, the neural network **400** is applied to the training input data to generate calculated output data. In particular, the training data and the calculated output data comprise a number of values, said number being equal with the number of nodes of the output layer.

[0056] In particular, a comparison between the calculated output data and the training data is used to recursively adapt

the weights within the neural network **400** (backpropagation algorithm). In particular, the weights are changed according to

$$w'^{(n)}_{i,j} = w^{(n)}_{i,j} - \gamma \cdot \delta^{(n)}_j \cdot x^{(n)}_i$$

wherein y is a learning rate, and the numbers $\delta^{(n)}_j$ can be recursively calculated as

$$\delta^{(n)}_j = \left(\sum_k \delta^{(n+1)}_k \cdot w^{(n+1)}_{j,k}\right) \cdot f'\left(\sum_i x^{(n)}_i \cdot w^{(n)}_{i,j}\right)$$

based on $\delta^{(n+1)}_j$, if the (n+1)-th layer is not the output layer, and

$$\delta^{(n)}_j = \left(x^{(n+1)}_j - t^{(n+1)}_j\right) \cdot f'\left(x^{(n)}_i \cdot w^{(n)}_{i,j}\right)$$

if the (n+1)-th layer is the output layer **413**, wherein f' is the first derivative of the activation function, and $t^{(n+1)}_j$ is the comparison training value for the j-th node of the output layer **413**.

[0057] A convolutional neural network is a neural network that uses a convolution operation instead of general matrix multiplication in at least one of its layers (so-called "convolutional layer"). In particular, a convolutional layer performs a dot product of one or more convolution kernels with the convolutional layer's input data/image, wherein the entries of the one or more convolution kernels are the parameters or weights that are adapted by training. In particular, one can use the Frobenius inner product and the ReLU activation function. A convolutional neural network can comprise additional layers, e.g., pooling layers, fully connected layers, and normalization layers.

[0058] By using convolutional neural networks input images can be processed in a very efficient way, because a convolution operation based on different kernels can extract various image features, so that by adapting the weights of the convolution kernel the relevant image features can be found during training. Furthermore, based on the weight-sharing in the convolutional kernels less parameters need to be trained, which prevents overfitting in the training phase and allows to have faster training or more layers in the network, improving the performance of the network.

[0059] FIG. **5** shows an embodiment of a convolutional neural network **500** that may be used to implement one or more machine learning models described herein. In the displayed embodiment, the convolutional neural network **500** comprises an input node layer **510**, a convolutional layer **511**, a pooling layer **513**, a fully connected layer **514** and an output node layer **516**, as well as hidden node layers **512**, **514**. Alternatively, the convolutional neural network **500** can comprise several convolutional layers **511**, several pooling layers **513** and several fully connected layers **515**, as well as other types of layers. The order of the layers can be chosen arbitrarily, usually fully connected layers **515** are used as the last layers before the output layer **516**.

[0060] In particular, within a convolutional neural network **500** nodes **520**, **522**, **524** of a node layer **510**, **512**, **514** can be considered to be arranged as a d-dimensional matrix or as a d-dimensional image. In particular, in the two-dimensional case the value of the node **520**, **522**, **524** indexed with i and j in the n-th node layer **510**, **512**, **514** can be denoted as x(n)[i, j]. However, the arrangement of the nodes **520**, **522**, **524** of one node layer **510**, **512**, **514** does not have an effect on the calculations executed within the convolutional neural network **500** as such, since these are given solely by the structure and the weights of the edges.

[0061] A convolutional layer **511** is a connection layer between an anterior node layer **510** (with node values x(n−1)) and a posterior node layer **512** (with node values x(n)). In particular, a convolutional layer **511** is characterized by the structure and the weights of the incoming edges forming a convolution operation based on a certain number of kernels. In particular, the structure and the weights of the edges of the convolutional layer **511** are chosen such that the values x(n) of the nodes **522** of the posterior node layer **512** are calculated as a convolution x(n)=K*x(n−1) based on the values x(n−1) of the nodes **520** anterior node layer **510**, where the convolution * is defined in the two-dimensional case as

$$x^{(n)}_k[i, j] = (K * x^{(n-1)})[i, j] = \sum_{i'} \sum_{j'} K[i', j'] \cdot x^{(n-1)}[i - i', j - j'].$$

[0062] Here the kernel K is a d-dimensional matrix (in this embodiment, a two-dimensional matrix), which is usually small compared to the number of nodes **520**, **522** (e.g., a 3×3 matrix, or a 5×5 matrix). In particular, this implies that the weights of the edges in the convolution layer **511** are not independent, but chosen such that they produce said convolution equation. In particular, for a kernel being a 3×3 matrix, there are only 9 independent weights (each entry of the kernel matrix corresponding to one independent weight), irrespectively of the number of nodes **520**, **522** in the anterior node layer **510** and the posterior node layer **512**.

[0063] In general, convolutional neural networks **500** use node layers **510**, **512**, **514** with a plurality of channels, in particular, due to the use of a plurality of kernels in convolutional layers **511**. In those cases, the node layers can be considered as (d+1)-dimensional matrices (the first dimension indexing the channels). The action of a convolutional layer **511** is then a two-dimensional example defined as

$$x^{(n)_b}[i, j] = \sum_a K_{a,b} * x^{(n-1)_a}[i, j] = \sum_a \sum_{i'} \sum_{j'} K_{a,b}[i', j'] \cdot x^{(n-1)_a}[i - i', j - j']$$

where $x^{(n-1)_a}$ corresponds to the a-th channel of the anterior node layer **510**, $x^{(n)_b}$ corresponds to the b-th channel of the posterior node layer **512** and $K_{a,b}$ corresponds to one of the kernels. If a convolutional layer **511** acts on an anterior node layer **510** with A channels and outputs a posterior node layer **512** with B channels, there are A·B independent d-dimensional kernels $K_{a,b}$.

[0064] In general, in convolutional neural networks **500** activation functions are used. In this embodiment ReLU (acronym for "Rectified Linear Units") is used, with R(z) =max(0, z), so that the action of the convolutional layer **511** in the two-dimensional example is

$$x^{(n)}{}_b[i,\ j] = R\Big(\sum_a \big(K_{a,b} * x^{(n-1)}{}_a\big)[i,\ j]\Big) = R\Big(\sum_a \sum_{i'} \sum_{j'} K_{a,b}[i',\ j'] \cdot x^{(n-1)}{}_a[i-i',\ j-j']\Big)$$

[0065] It is also possible to use other activation functions, e.g., ELU (acronym for "Exponential Linear Unit"), LeakyReLU, Sigmoid, Tanh or Softmax.

[0066] In the displayed embodiment, the input layer **510** comprises 36 nodes **520**, arranged as a two-dimensional 6×6 matrix. The first hidden node layer **512** comprises 72 nodes **522**, arranged as two two-dimensional 6×6 matrices, each of the two matrices being the result of a convolution of the values of the input layer with a 3×3 kernel within the convolutional layer **511**. Equivalently, the nodes **522** of the first hidden node layer **512** can be interpreted as arranged as a three-dimensional 2×6×6 matrix, wherein the first dimension correspond to the channel dimension.

[0067] The advantage of using convolutional layers **511** is that spatially local correlation of the input data can exploited by enforcing a local connectivity pattern between nodes of adjacent layers, in particular by each node being connected to only a small region of the nodes of the preceding layer.

[0068] A pooling layer **513** is a connection layer between an anterior node layer **512** (with node values x(n−1)) and a posterior node layer **514** (with node values x(n)). In particular, a pooling layer **513** can be characterized by the structure and the weights of the edges and the activation function forming a pooling operation based on a non-linear pooling function f. For example, in the two-dimensional case the values x(n) of the nodes **524** of the posterior node layer **514** can be calculated based on the values x(n−1) of the nodes **522** of the anterior node layer **512** as

$$x^{(n)}{}_b[i,\ j] = f\big(x^{(n-1)}[id_1,\ jd_2],\ \ldots,\ x^{(n-1)}{}_b[(i+1)d_1-1,\ (j+1)d_2-1]\big)$$

[0069] In other words, by using a pooling layer **513** the number of nodes **522**, **524** can be reduced, by re-placing a number d1·d2 of neighboring nodes **522** in the anterior node layer **512** with a single node **522** in the posterior node layer **514** being calculated as a function of the values of said number of neighboring nodes. In particular, the pooling function f can be the max-function, the average or the L2-Norm. In particular, for a pooling layer **513** the weights of the incoming edges are fixed and are not modified by training.

[0070] The advantage of using a pooling layer **513** is that the number of nodes **522**, **524** and the number of parameters is reduced. This leads to the amount of computation in the network being reduced and to a control of overfitting.

[0071] In the displayed embodiment, the pooling layer **513** is a max-pooling layer, replacing four neighboring nodes with only one node, the value being the maximum of the values of the four neighboring nodes. The max-pooling is applied to each d-dimensional matrix of the previous layer; in this embodiment, the max-pooling is applied to each of the two two-dimensional matrices, reducing the number of nodes from 72 to 18.

[0072] In general, the last layers of a convolutional neural network **500** are fully connected layers **515**. A fully connected layer **515** is a connection layer between an anterior node layer **514** and a posterior node layer **516**. A fully connected layer **513** can be characterized by the fact that a majority, in particular, all edges between nodes **514** of the anterior node layer **514** and the nodes **516** of the posterior node layer are present, and wherein the weight of each of these edges can be adjusted individually.

[0073] In this embodiment, the nodes **524** of the anterior node layer **514** of the fully connected layer **515** are displayed both as two-dimensional matrices, and additionally as non-related nodes (indicated as a line of nodes, wherein the number of nodes was reduced for a better presentability). This operation is also denoted as "flattening". In this embodiment, the number of nodes **526** in the posterior node layer **516** of the fully connected layer **515** smaller than the number of nodes **524** in the anterior node layer **514**. Alternatively, the number of nodes **526** can be equal or larger.

[0074] Furthermore, in this embodiment the Softmax activation function is used within the fully connected layer **515**. By applying the Softmax function, the sum the values of all nodes **526** of the output layer **516** is 1, and all values of all nodes **526** of the output layer **516** are real numbers between 0 and 1. In particular, if using the convolutional neural network **500** for categorizing input data, the values of the output layer **516** can be interpreted as the probability of the input data falling into one of the different categories.

[0075] In particular, convolutional neural networks **500** can be trained based on the backpropagation algorithm. For preventing overfitting, methods of regularization can be used, e.g., dropout of nodes **520**, . . . , **524**, stochastic pooling, use of artificial data, weight decay based on the L1 or the L2 norm, or max norm constraints.

[0076] According to an aspect, the machine learning model may comprise one or more residual networks (ResNet). In particular, a ResNet is an artificial neural network comprising at least one jump or skip connection used to jump over at least one layer of the artificial neural network. In particular, a ResNet may be a convolutional neural network comprising one or more skip connections respectively skipping one or more convolutional layers. According to some examples, the ResNets may be represented as m-layer ResNets, where m is the number of layers in the corresponding architecture and, according to some examples, may take values of 34, 50, 101, or 152. According to some examples, such an m-layer ResNet may respectively comprise (m−2)/2 skip connections.

[0077] A skip connection may be seen as a bypass which directly feeds the output of one preceding layer over one or more bypassed layers to a layer succeeding the one or more bypassed layers. Instead of having to directly fit a desired mapping, the bypassed layers would then have to fit a residual mapping "balancing" the directly fed output.

[0078] Fitting the residual mapping is computationally easier to optimize than the directed mapping. What is more, this alleviates the problem of vanishing/exploding gradients during optimization upon training the machine learning models: if a bypassed layer runs into such problems, its contribution may be skipped by regularization of the directly fed output. Using ResNets thus brings about the advantage that much deeper networks may be trained.

[0079] A generative adversarial model (an acronym is GA model) comprises a generative function and a discriminative function, wherein the generative function creates synthetic data, and the discriminative function distinguishes between synthetic and real data. By training the generative function

and/or the discriminative function on the one hand the generative function is configured to create synthetic data which is incorrectly classified by the discriminative function as real, on the other hand the discriminative function is configured to distinguish between real data and synthetic data generated by the generative function. In the notion of game theory, a generative adversarial model can be interpreted as a zero-sum game. The training of the generative function and/or of the discriminative function is based, in particular, on the minimization of a cost function.

[0080] By using a GA model, based on a set of training data synthetic data can be generated that has the same characteristics as the training data set. The training of the GA model can be based on data not being annotated (unsupervised learning), so that there is low effort in training a GA model.

[0081] FIG. **6** shows a data flow diagram according to an embodiment for using a generative adversarial network for creating synthetic output data G(x) **608** based on input data x **602** that is indistinguishable from real output data y **604**, in accordance with one or more embodiments. The synthetic output data G(x) **608** has the same structure as the real output data y **604**, but its content is not derived from real world data.

[0082] The generative adversarial network comprises a generator function G **606** and a classifier function C **610** which are trained jointly. The task of the generator function G **606** is to provide realistic synthetic output data G(x) **608** based on input data x **602**, and the task of the classifier function C **610** is to distinguish between real output data y **604** and synthetic output data G(x) **608**. In particular, the output of the classifier function C **610** is a real number between 0 and 1 corresponding to the probability of the input value being real data, so that an ideal classifier function would calculate an output value of C(y) **614**≈1 for real data y **604** and C(G(x)) **612**≈0 for synthetic data G(x) **608**.

[0083] Within the training process, parameters of the generator function G **606** are adapted so that the synthetic output data G(x) **608** has the same characteristics as real output data y **604**, so that the classifier function C **610** cannot distinguish between real and synthetic data anymore. At the same time, parameters of the classifier function C **610** are adapted so that it distinguishes between real and synthetic data in the best possible way. Here, the training relies on pairs comprising input data x **602** and the corresponding real output data y **604**. Within a single training step, the generator function G **606** is applied to the input data x **602** for generating synthetic output data G(x) **608**. Furthermore, the classifier function C **610** is applied to the real output data y **604** for generating a first classification result C(y) **614**. Additionally, the classifier function C **610** is applied to the synthetic output data G(x) **608** for generating a second classification result C(G(x)) **612**.

[0084] Adapting the parameters of the generative function G **606** and the classifier function C **610** is based on minimizing a cost function by using the backpropagation algorithm, respectively. In this embodiment, the cost function K_C for the classifier function C **610** is $K_C \propto -\mathrm{BCE}(C(y), 1) - \mathrm{BCE}(C(G(x)), 0)$, wherein BCE denotes the binary cross entropy defined as $\mathrm{BCE}(z, z') = z' \cdot \log(z) + (1 - z') \cdot \log(1 - z)$. By using this cost function, both wrongly classifying real output data as synthetic (indicated by C(y)≈0) and wrongly classifying synthetic output data as real (indicated as C(G(x)) **612**≈1) increases the cost function K_C to be minimized. Furthermore, the cost function K_G for the generator function G **606** is $K_G \propto -\mathrm{BCE}(C(G(x)), 1) = -\log(C(G(x)))$. By using this cost function, correctly classified synthetic output data (indicated as C(G(x)) **612**≈0) leads to an increase of the cost function K_G to be minimized.

[0085] In particular, a recurrent machine learning model is a machine learning model whose output does not only depend on the input value and the parameters of the machine learning model adapted by the training process, but also on a hidden state vector, wherein the hidden state vector is based on previous inputs used on for the recurrent machine learning model. In particular, the recurrent machine learning model can comprise additional storage states or additional structures that incorporate time delays or comprise feedback loops.

[0086] In particular, the underlying structure of a recurrent machine learning model can be a neural network, which can be denoted as recurrent neural network. Such a recurrent neural network can be described as an artificial neural network where connections between nodes form a directed graph along a temporal sequence. In particular, a recurrent neural network can be interpreted as directed acyclic graph. In particular, the recurrent neural network can be a finite impulse recurrent neural network or an infinite impulse recurrent neural network (wherein a finite impulse network can be unrolled and replaced with a strictly feedforward neural network, and an infinite impulse network cannot be unrolled and replaced with a strictly feedforward neural network).

[0087] In particular, training a recurrent neural network can be based on the BPTT algorithm (acronym for "back-propagation through time"), on the RTRL algorithm (acronym for "real-time recurrent learning") and/or on genetic algorithms.

[0088] By using a recurrent machine learning model input data comprising sequences of variable length can be used. In particular, this implies that the method cannot be used only for a fixed number of input datasets (and needs to be trained differently for every other number of input datasets used as input), but can be used for an arbitrary number of input datasets. This implies that the whole set of training data, independent of the number of input datasets contained in different sequences, can be used within the training, and that training data is not reduced to training data corresponding to a certain number of successive input datasets.

[0089] FIG. **7** shows the schematic structure of a recurrent machine learning model F, both in a recurrent representation **702** and in an unfolded representation **704**, that may be used to implement one or more machine learning models described herein. The recurrent machine learning model takes as input several input datasets x, $x_1, \ldots, x_N$ **706** and creates a corresponding set of output datasets y, $y_1, \ldots, y_N$ **708**. Furthermore, the output depends on a so-called hidden vector h, $h_1, \ldots, h_N$ **710**, which implicitly comprises information about input datasets previously used as input for the recurrent machine learning model F **712**. By using these hidden vectors h, $h_1, \ldots, h_N$ **710**, a sequentiality of the input datasets can be leveraged.

[0090] In a single step of the processing, the recurrent machine learning model F **712** takes as input the hidden vector h_{n-1} created within the previous step and an input dataset x_n. Within this step, the recurrent machine learning model F generates as output an updated hidden vector ha and an output dataset y_n. In other words, one step of processing

calculates $(y_n, h_n)=F(x_n, h_{n-1})$, or by splitting the recurrent machine learning model F **712** into a part F(y) calculating the output data and F(h) calculating the hidden vector, one step of processing calculates $y_n=F^{(y)}(x_n, h_{n-1})$ and $h_n=F^{(h)}(x_n, h_{n-1})$. For the first processing step, h_0 can be chosen randomly or filled with all entries being zero. The parameters of the recurrent machine learning model F **712** that were trained based on training datasets before do not change between the different processing steps.

[0091] In particular, the output data and the hidden vector of a processing step depend on all the previous input datasets used in the previous steps. $y_n=F(^y)(x_n, F^{(h)}(x_{n-1}, h_{n-2}))$ and $h_n=F(h)(x_n, F^{(h)}(x_{n-1}, h_{n-2}))$.

[0092] Systems, apparatuses, and methods described herein may be implemented using digital circuitry, or using one or more computers using well-known computer processors, memory units, storage devices, computer software, and other components. Typically, a computer includes a processor for executing instructions and one or more memories for storing instructions and data. A computer may also include, or be coupled to, one or more mass storage devices, such as one or more magnetic disks, internal hard disks and removable disks, magneto-optical disks, optical disks, etc.

[0093] Systems, apparatuses, and methods described herein may be implemented using computers operating in a client-server relationship. Typically, in such a system, the client computers are located remotely from the server computer and interact via a network. The client-server relationship may be defined and controlled by computer programs running on the respective client and server computers.

[0094] Systems, apparatuses, and methods described herein may be implemented within a network-based cloud computing system. In such a network-based cloud computing system, a server or another processor that is connected to a network communicates with one or more client computers via a network. A client computer may communicate with the server via a network browser application residing and operating on the client computer, for example. A client computer may store data on the server and access the data via the network. A client computer may transmit requests for data, or requests for online services, to the server via the network. The server may perform requested services and provide data to the client computer(s). The server may also transmit data adapted to cause a client computer to perform a specified function, e.g., to perform a calculation, to display specified data on a screen, etc. For example, the server may transmit a request adapted to cause a client computer to perform one or more of the steps or functions of the methods and workflows described herein, including one or more of the steps or functions of FIG. **1** or **2**. Certain steps or functions of the methods and workflows described herein, including one or more of the steps or functions of FIG. **1** or **2**, may be performed by a server or by another processor in a network-based cloud-computing system. Certain steps or functions of the methods and workflows described herein, including one or more of the steps of FIG. **1** or **2**, may be performed by a client computer in a network-based cloud computing system. The steps or functions of the methods and workflows described herein, including one or more of the steps of FIG. **1** or **2**, may be performed by a server and/or by a client computer in a network-based cloud computing system, in any combination.

[0095] Systems, apparatuses, and methods described herein may be implemented using a computer program product tangibly embodied in an information carrier, e.g., in a non-transitory machine-readable storage device, for execution by a programmable processor; and the method and workflow steps described herein, including one or more of the steps or functions of FIG. **1** or **2**, may be implemented using one or more computer programs that are executable by such a processor. A computer program is a set of computer program instructions that can be used, directly or indirectly, in a computer to perform a certain activity or bring about a certain result. A computer program can be written in any form of programming language, including compiled or interpreted languages, and it can be deployed in any form, including as a stand-alone program or as a module, component, subroutine, or other unit suitable for use in a computing environment.

[0096] A high-level block diagram of an example computer **802** that may be used to implement systems, apparatuses, and methods described herein is depicted in FIG. **8**. Computer **802** includes a processor **804** operatively coupled to a data storage device **812** and a memory **810**. Processor **804** controls the overall operation of computer **802** by executing computer program instructions that define such operations. The computer program instructions may be stored in data storage device **812**, or other computer readable medium, and loaded into memory **810** when execution of the computer program instructions is desired. Thus, the method and workflow steps or functions of FIG. **1** or **2** can be defined by the computer program instructions stored in memory **810** and/or data storage device **812** and controlled by processor **804** executing the computer program instructions. For example, the computer program instructions can be implemented as computer executable code programmed by one skilled in the art to perform the method and workflow steps or functions of FIG. **1** or **2**. Accordingly, by executing the computer program instructions, the processor **804** executes the method and workflow steps or functions of FIG. **1** or **2**. Computer **802** may also include one or more network interfaces **806** for communicating with other devices via a network. Computer **802** may also include one or more input/output devices **808** that enable user interaction with computer **802** (e.g., display, keyboard, mouse, speakers, buttons, etc.).

[0097] Processor **804** may include both general and special purpose microprocessors, and may be the sole processor or one of multiple processors of computer **802**. Processor **804** may include one or more central processing units (CPUs), for example. Processor **804**, data storage device **812**, and/or memory **810** may include, be supplemented by, or incorporated in, one or more application-specific integrated circuits (ASICs) and/or one or more field programmable gate arrays (FPGAs).

[0098] Data storage device **812** and memory **810** each include a tangible non-transitory computer readable storage medium. Data storage device **812**, and memory **810**, may each include high-speed random access memory, such as dynamic random access memory (DRAM), static random access memory (SRAM), double data rate synchronous dynamic random access memory (DDR RAM), or other random access solid state memory devices, and may include non-volatile memory, such as one or more magnetic disk storage devices such as internal hard disks and removable disks, magneto-optical disk storage devices, optical disk storage devices, flash memory devices, semiconductor memory devices, such as erasable programmable read-only

memory (EPROM), electrically erasable programmable read-only memory (EEPROM), compact disc read-only memory (CD-ROM), digital versatile disc read-only memory (DVD-ROM) disks, or other non-volatile solid state storage devices.

[0099] Input/output devices **808** may include peripherals, such as a printer, scanner, display screen, etc. For example, input/output devices **808** may include a display device such as a cathode ray tube (CRT) or liquid crystal display (LCD) monitor for displaying information to the user, a keyboard, and a pointing device such as a mouse or a trackball by which the user can provide input to computer **802**.

[0100] An image acquisition device **814** can be connected to the computer **802** to input image data (e.g., medical images) to the computer **802**. It is possible to implement the image acquisition device **814** and the computer **802** as one device. It is also possible that the image acquisition device **814** and the computer **802** communicate wirelessly through a network. In a possible embodiment, the computer **802** can be located remotely with respect to the image acquisition device **814**.

[0101] Any or all of the systems, apparatuses, and methods discussed herein may be implemented using one or more computers such as computer **802**.

[0102] One skilled in the art will recognize that an implementation of an actual computer or computer system may have other structures and may contain other components as well, and that FIG. **8** is a high level representation of some of the components of such a computer for illustrative purposes.

[0103] Independent of the grammatical term usage, individuals with male, female or other gender identities are included within the term.

[0104] The foregoing Detailed Description is to be understood as being in every respect illustrative and exemplary, but not restrictive, and the scope of the invention disclosed herein is not to be determined from the Detailed Description, but rather from the claims as interpreted according to the full breadth permitted by the patent laws. It is to be understood that the embodiments shown and described herein are only illustrative of the principles of the present invention and that various modifications may be implemented by those skilled in the art without departing from the scope and spirit of the invention. Those skilled in the art could implement various other feature combinations without departing from the scope and spirit of the invention.

[0105] The following is a list of non-limiting illustrative embodiments disclosed herein:

[0106] Illustrative embodiment 1. A computer-implemented method comprising: receiving 1) a plurality of medical images acquired at a plurality of acquisition orientations and in one or more domains and 2) a domain code for each particular acquisition orientation of the plurality of acquisition orientations, each of the domain codes identifying a presence of the one or more domains of the plurality of medical images that were acquired at the particular acquisition orientation; for each of the particular acquisition orientations: encoding the domain code for the particular acquisition orientation, extracting features from the plurality of medical images that were acquired at the particular acquisition orientation, and combining the encoded domain code and the extracted features to generate image features for the particular acquisition orientation; performing a medical imaging analysis task based on the image features for each of the particular acquisition orientations; and outputting results of the medical imaging analysis task.

[0107] Illustrative embodiment 2. The computer-implemented method of illustrative embodiment 1, wherein extracting features from the plurality of medical images that were acquired at the particular acquisition orientation comprises: extracting a first set of features from the plurality of medical images that were acquired at the particular acquisition orientation using a first machine learning based encoder; and encoding the first set of features into a second set of features using a second machine learning based encoder to generate the extracted features.

[0108] Illustrative embodiment 3. The computer-implemented method of illustrative embodiment 2, wherein extracting a first set of features from the plurality of medical images that were acquired at the particular acquisition orientation using a first machine learning based encoder comprises: encoding the first set of features with positional embeddings.

[0109] Illustrative embodiment 4. The computer-implemented method of any one of illustrative embodiments 2-3, wherein the first set of features are lower level features and the second set of features are higher level features.

[0110] Illustrative embodiment 5. The computer-implemented method of any one of illustrative embodiments 2-4, wherein extracting a first set of features from the plurality of medical images that were acquired at the particular acquisition orientation using a first machine learning based encoder comprises: resampling the plurality of medical images that were acquired at the particular acquisition orientation onto a same pixel grid for that particular acquisition orientation to form aligned images; combining the aligned images to form an input tensor; and extracting the first set of features from the input tensor using the first machine learning based encoder.

[0111] Illustrative embodiment 6. The computer-implemented method of any one of illustrative embodiments 1-5, wherein combining the encoded domain code and the extracted features to generate image features for the particular acquisition orientation comprises: combining the encoded domain code and the extracted features using a cross-attention layer.

[0112] Illustrative embodiment 7. The computer-implemented method of any one of illustrative embodiments 1-6, wherein performing a medical imaging analysis task based on the image features for each of the particular acquisition orientations comprises: combining the image features for each of the particular acquisition orientations to generate combined features; and performing the medical imaging analysis task based on the combined features.

[0113] Illustrative embodiment 8. The computer-implemented method of any one of illustrative embodiments 1-7, wherein each of the domain codes identify an absence of certain domains of a set of predefined domains from the one or more domains of the plurality of medical images that were acquired at the particular acquisition orientation.

[0114] Illustrative embodiment 9. The computer-implemented method of any one of illustrative embodiments 1-8, wherein the plurality of acquisition orientations comprises at least one of axial, sagittal, or coronal.

[0115] Illustrative embodiment 10. An apparatus comprising: means for receiving 1) a plurality of medical images acquired at a plurality of acquisition orientations and in one or more domains and 2) a domain code for each particular

acquisition orientation of the plurality of acquisition orientations, each of the domain codes identifying a presence of the one or more domains of the plurality of medical images that were acquired at the particular acquisition orientation; for each of the particular acquisition orientations: means for encoding the domain code for the particular acquisition orientation, means for extracting features from the plurality of medical images that were acquired at the particular acquisition orientation, and means for combining the encoded domain code and the extracted features to generate image features for the particular acquisition orientation; means for performing a medical imaging analysis task based on the image features for each of the particular acquisition orientations; and means for outputting results of the medical imaging analysis task.

[0116] Illustrative embodiment 11. The apparatus of illustrative embodiment 10, wherein the means for extracting features from the plurality of medical images that were acquired at the particular acquisition orientation comprises: means for extracting a first set of features from the plurality of medical images that were acquired at the particular acquisition orientation using a first machine learning based encoder; and means for encoding the first set of features into a second set of features using a second machine learning based encoder to generate the extracted features.

[0117] Illustrative embodiment 12. The apparatus of any one of illustrative embodiments 10-11, wherein the means for extracting a first set of features from the plurality of medical images that were acquired at the particular acquisition orientation using a first machine learning based encoder comprises: means for encoding the first set of features with positional embeddings.

[0118] Illustrative embodiment 13. The apparatus of any one of illustrative embodiments 10-12, wherein the first set of features are lower level features and the second set of features are higher level features.

[0119] Illustrative embodiment 14. The apparatus of any one of illustrative embodiments 10-13, wherein the means for extracting a first set of features from the plurality of medical images that were acquired at the particular acquisition orientation using a first machine learning based encoder comprises: means for resampling the plurality of medical images that were acquired at the particular acquisition orientation onto a same pixel grid for that particular acquisition orientation to form aligned images; means for combining the aligned images to form an input tensor; and means for extracting the first set of features from the input tensor using the first machine learning based encoder.

[0120] Illustrative embodiment 15. A non-transitory computer-readable storage medium comprising instructions which, when executed by a computer, cause the computer to carry out operations comprising: receiving 1) a plurality of medical images acquired at a plurality of acquisition orientations and in one or more domains and 2) a domain code for each particular acquisition orientation of the plurality of acquisition orientations, each of the domain codes identifying a presence of the one or more domains of the plurality of medical images that were acquired at the particular acquisition orientation; for each of the particular acquisition orientations: encoding the domain code for the particular acquisition orientation, extracting features from the plurality of medical images that were acquired at the particular acquisition orientation, and combining the encoded domain code and the extracted features to generate image features for the particular acquisition orientation; performing a medical imaging analysis task based on the image features for each of the particular acquisition orientations; and outputting results of the medical imaging analysis task.

[0121] Illustrative embodiment 16. The non-transitory computer-readable storage medium of illustrative embodiment 15, wherein extracting features from the plurality of medical images that were acquired at the particular acquisition orientation comprises: extracting a first set of features from the plurality of medical images that were acquired at the particular acquisition orientation using a first machine learning based encoder; and encoding the first set of features into a second set of features using a second machine learning based encoder to generate the extracted features.

[0122] Illustrative embodiment 17. The non-transitory computer-readable storage medium of any one of illustrative embodiments 15-16, wherein combining the encoded domain code and the extracted features to generate image features for the particular acquisition orientation comprises: combining the encoded domain code and the extracted features using a cross-attention layer.

[0123] Illustrative embodiment 18. The non-transitory computer-readable storage medium of any one of illustrative embodiments 15-17, wherein performing a medical imaging analysis task based on the image features for each of the particular acquisition orientations comprises: combining the image features for each of the particular acquisition orientations to generate combined features; and performing the medical imaging analysis task based on the combined features.

[0124] Illustrative embodiment 19. The non-transitory computer-readable storage medium of any one of illustrative embodiments 15-18, wherein each of the domain codes identify an absence of certain domains of a set of predefined domains from the one or more domains of the plurality of medical images that were acquired at the particular acquisition orientation.

[0125] Illustrative embodiment 20. The non-transitory computer-readable storage medium of any one of illustrative embodiments 15-19, wherein the plurality of acquisition orientations comprises at least one of axial, sagittal, or coronal.

1. A computer-implemented method comprising:

receiving 1) a plurality of medical images acquired at a plurality of acquisition orientations and in one or more domains and 2) a domain code for each particular acquisition orientation of the plurality of acquisition orientations, each of the domain codes identifying a presence of the one or more domains of the plurality of medical images that were acquired at the particular acquisition orientation;

for each of the particular acquisition orientations:

encoding the domain code for the particular acquisition orientation,

extracting features from the plurality of medical images that were acquired at the particular acquisition orientation, and

combining the encoded domain code and the extracted features to generate image features for the particular acquisition orientation;

performing a medical imaging analysis task based on the image features for each of the particular acquisition orientations; and

outputting results of the medical imaging analysis task.

2. The computer-implemented method of claim 1, wherein extracting features from the plurality of medical images that were acquired at the particular acquisition orientation comprises:

extracting a first set of features from the plurality of medical images that were acquired at the particular acquisition orientation using a first machine learning based encoder; and

encoding the first set of features into a second set of features using a second machine learning based encoder to generate the extracted features.

3. The computer-implemented method of claim 2, wherein extracting a first set of features from the plurality of medical images that were acquired at the particular acquisition orientation using a first machine learning based encoder comprises:

encoding the first set of features with positional embeddings.

4. The computer-implemented method of claim 2, wherein the first set of features are lower level features and the second set of features are higher level features.

5. The computer-implemented method of claim 2, wherein extracting a first set of features from the plurality of medical images that were acquired at the particular acquisition orientation using a first machine learning based encoder comprises:

resampling the plurality of medical images that were acquired at the particular acquisition orientation onto a same pixel grid for that particular acquisition orientation to form aligned images;

combining the aligned images to form an input tensor; and

extracting the first set of features from the input tensor using the first machine learning based encoder.

6. The computer-implemented method of claim 1, wherein combining the encoded domain code and the extracted features to generate image features for the particular acquisition orientation comprises:

combining the encoded domain code and the extracted features using a cross-attention layer.

7. The computer-implemented method of claim 1, wherein performing a medical imaging analysis task based on the image features for each of the particular acquisition orientations comprises:

combining the image features for each of the particular acquisition orientations to generate combined features; and

performing the medical imaging analysis task based on the combined features.

8. The computer-implemented method of claim 1, wherein each of the domain codes identify an absence of certain domains of a set of predefined domains from the one or more domains of the plurality of medical images that were acquired at the particular acquisition orientation.

9. The computer-implemented method of claim 1, wherein the plurality of acquisition orientations comprises at least one of axial, sagittal, or coronal.

10. An apparatus comprising:

means for receiving 1) a plurality of medical images acquired at a plurality of acquisition orientations and in one or more domains and 2) a domain code for each particular acquisition orientation of the plurality of acquisition orientations, each of the domain codes identifying a presence of the one or more domains of the plurality of medical images that were acquired at the particular acquisition orientation;

for each of the particular acquisition orientations:

means for encoding the domain code for the particular acquisition orientation,

means for extracting features from the plurality of medical images that were acquired at the particular acquisition orientation, and

means for combining the encoded domain code and the extracted features to generate image features for the particular acquisition orientation;

means for performing a medical imaging analysis task based on the image features for each of the particular acquisition orientations; and

means for outputting results of the medical imaging analysis task.

11. The apparatus of claim 10, wherein the means for extracting features from the plurality of medical images that were acquired at the particular acquisition orientation comprises:

means for extracting a first set of features from the plurality of medical images that were acquired at the particular acquisition orientation using a first machine learning based encoder; and

means for encoding the first set of features into a second set of features using a second machine learning based encoder to generate the extracted features.

12. The apparatus of claim 11, wherein the means for extracting a first set of features from the plurality of medical images that were acquired at the particular acquisition orientation using a first machine learning based encoder comprises:

means for encoding the first set of features with positional embeddings.

13. The apparatus of claim 11, wherein the first set of features are lower level features and the second set of features are higher level features.

14. The apparatus of claim 11, wherein the means for extracting a first set of features from the plurality of medical images that were acquired at the particular acquisition orientation using a first machine learning based encoder comprises:

means for resampling the plurality of medical images that were acquired at the particular acquisition orientation onto a same pixel grid for that particular acquisition orientation to form aligned images;

means for combining the aligned images to form an input tensor; and

means for extracting the first set of features from the input tensor using the first machine learning based encoder.

15. A non-transitory computer-readable storage medium comprising instructions which, when executed by a computer, cause the computer to carry out operations comprising:

receiving 1) a plurality of medical images acquired at a plurality of acquisition orientations and in one or more domains and 2) a domain code for each particular acquisition orientation of the plurality of acquisition orientations, each of the domain codes identifying a presence of the one or more domains of the plurality of medical images that were acquired at the particular acquisition orientation;

for each of the particular acquisition orientations:

encoding the domain code for the particular acquisition orientation,

extracting features from the plurality of medical images that were acquired at the particular acquisition orientation, and

combining the encoded domain code and the extracted features to generate image features for the particular acquisition orientation;

performing a medical imaging analysis task based on the image features for each of the particular acquisition orientations; and

outputting results of the medical imaging analysis task.

16. The non-transitory computer-readable storage medium of claim **15**, wherein extracting features from the plurality of medical images that were acquired at the particular acquisition orientation comprises:

extracting a first set of features from the plurality of medical images that were acquired at the particular acquisition orientation using a first machine learning based encoder; and

encoding the first set of features into a second set of features using a second machine learning based encoder to generate the extracted features.

17. The non-transitory computer-readable storage medium of claim **15**, wherein combining the encoded domain code and the extracted features to generate image features for the particular acquisition orientation comprises:

combining the encoded domain code and the extracted features using a cross-attention layer.

18. The non-transitory computer-readable storage medium of claim **15**, wherein performing a medical imaging analysis task based on the image features for each of the particular acquisition orientations comprises:

combining the image features for each of the particular acquisition orientations to generate combined features; and

performing the medical imaging analysis task based on the combined features.

19. The non-transitory computer-readable storage medium of claim **15**, wherein each of the domain codes identify an absence of certain domains of a set of predefined domains from the one or more domains of the plurality of medical images that were acquired at the particular acquisition orientation.

20. The non-transitory computer-readable storage medium of claim **15**, wherein the plurality of acquisition orientations comprises at least one of axial, sagittal, or coronal.

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